Idea of Gaussian Elimination Method

1) Swapping two equations (rows) doesn't change the system.

2) Multiplying an equation (row) by a non-zero constant doesn't change the system.

3) Adding or subtracting two equations (rows) and replacing one of them with the resulting equation doesn't change the system.

Row echelon form: Our Target:

$$\left(\begin{array}{ccc|c} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{array}\right) \sim \left(\begin{array}{ccc|c} \textcircled{1} & \cdot & \cdot & \cdot \\ 0 & \textcircled{1} & \cdot & \cdot \\ 0 & 0 & \textcircled{1} & \cdot \end{array}\right)$$

① = pivot/leader

Problem-1: Convert the system into Row Echelon Form and solve by Back Substitution.

$$x + y + 2z = 9$$
$$2x + 4y - 3z = 1$$
$$3x + 6y - 5z = 0$$

CamScanner

Solution: $x+y+2z=9$

$2x+4y-3z=1$

$3x+6y-5z=0$

$$\sim \left(\begin{array}{ccc|c} \textcircled{1} & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{array}\right)$$ [Augmented Matrix]

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & \textcircled{2} & -7 & -17 \\ 0 & 3 & -11 & -27 \end{array}\right)$$ $[R_2' = R_2 - 2R_1,\ R_3' = R_3 - 3R_1]$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 0 & -1 & -3 \end{array}\right)$$ $[R_3' = 2R_3 - 3R_2]$

Then the original system:

$x+y+2z=9$

$2y-7z=-17$

$-z=-3$

$\therefore z=3$

$2y-7(3)=-17$

$\Rightarrow 2y=-17+21$

$\therefore y=2.$

$x+2+2(3)=9$

$\therefore x=1.$

$\therefore (x,y,z)=(1,2,3).$

Unique Solution.

CamScanner

*Reduced Row Echelon Form:

$$\left(\begin{array}{ccc|c} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & \cdot \\ 0 & 1 & 0 & \cdot \\ 0 & 0 & 1 & \cdot \end{array}\right) \rightarrow$$ Reduced Row Echelon Form.

2) Convert the system into Reduced Row Echelon Form.

$$2x - y + z = -3$$
$$4x - 2y + z = -8$$
$$x + 3y - 2z = 4$$

$$\sim \left(\begin{array}{ccc|c} 2 & -1 & 1 & -3 \\ 4 & -2 & 1 & -8 \\ 1 & 3 & -2 & 4 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} 2 & -1 & 1 & -3 \\ 0 & 0 & -1 & -2 \\ 0 & 7 & -5 & 11 \end{array}\right) \quad \begin{bmatrix} R_2' = R_2 - 2R_1 \\ R_3' = 2R_3 - R_1 \end{bmatrix}$$

$$\sim \left(\begin{array}{ccc|c} 2 & -1 & 1 & -3 \\ 0 & 7 & -5 & 11 \\ 0 & 0 & -1 & -2 \end{array}\right) \quad [R_2 \leftrightarrow R_3]$$

[Row Echelon Form]

$$\sim \left(\begin{array}{ccc|c} 2 & -1 & 0 & -5 \\ 0 & 7 & 0 & 21 \\ 0 & 0 & -1 & -2 \end{array}\right) \quad \begin{bmatrix} R_2' = R_2 - 5R_3 \\ R_1' = R_1 + R_3 \end{bmatrix}$$

CamScanner

$$\sim \left(\begin{array}{ccc|c} 14 & 0 & 0 & -14 \\ 0 & 7 & 0 & 21 \\ 0 & 0 & -1 & -2 \end{array}\right) \quad [R_1' = 7R_1 + R_2]$$

$$\sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 2 \end{array}\right) \quad \begin{bmatrix} R_1' = \frac{1}{14}R_1 \\ R_2' = \frac{1}{7}R_2 \\ R_3' = -R_3 \end{bmatrix}$$

Reduced Row Echelon Form.

Original System: $x = -1, y = 3, z = 2$.

$(x, y, z) = (-1, 3, 2)$

Problem-3: Convert the system into Reduced Row Echelon Form and solve it.

$$2x - 3y + 5z = 1$$
$$3x + y - z = 2$$
$$x + 4y - 6z = 1$$

$$\therefore \sim \left(\begin{array}{ccc|c} 2 & -3 & 5 & 1 \\ 3 & 1 & -1 & 2 \\ 1 & 4 & -6 & 1 \end{array}\right) \quad \begin{bmatrix} \cancel{R_2' = 2R_2 - 3R_1} \\ \cancel{R_3' = 2R_3 - R_1} \end{bmatrix}$$

$$\therefore \sim \left(\begin{array}{ccc|c} \textcircled{2} & -3 & 5 & 1 \\ 0 & 11 & -17 & 1 \\ 0 & 11 & -17 & 1 \end{array}\right) \quad \begin{bmatrix} R_2' = 2R_2 - 3R_1 \\ R_3' = 2R_3 - R_1 \end{bmatrix}$$

CamScanner

Problem-4: Convert the system into Reduced Row Echelon Form and solve the equation.

Solution: $x-4y+5z=8$

$3x+7y-z=3$

$x+15y-11z=-14$

$$\sim\left(\begin{array}{ccc|c} 1 & -4 & 5 & 8 \\ 3 & 7 & -1 & 3 \\ 1 & 15 & -11 & -14 \end{array}\right)$$

$$\sim\left(\begin{array}{ccc|c} 1 & -4 & 5 & 8 \\ 0 & 19 & -16 & -21 \\ 0 & 19 & -16 & -22 \end{array}\right) \quad \left[\begin{array}{l} R_2 = R_2 - 3R_1 \\ R_3 = R_3 - R_1 \end{array}\right]$$

$$\sim\left(\begin{array}{ccc|c} 1 & -4 & 5 & 8 \\ 0 & 19 & -16 & -21 \\ 0 & 0 & 0 & -1 \end{array}\right) \quad [R_3' = R_3 - R_2]$$

$$\sim\left(\begin{array}{ccc|c} 19 & 0 & 31 & 68 \\ 0 & 19 & -16 & -21 \\ 0 & 0 & 0 & -1 \end{array}\right) \quad [R_1' = 19R_1 + 4R_2]$$

$$\sim\left(\begin{array}{ccc|c} 1 & 0 & \frac{31}{19} & \frac{68}{19} \\ 0 & 1 & -\frac{16}{19} & -\frac{21}{19} \\ 0 & 0 & 0 & -1 \end{array}\right) \quad [\text{Reduced Row Echelon Form}]$$

Original System: $x + \frac{31}{19}z = \frac{68}{19}$

$y - \frac{16}{19} = -\frac{21}{19}$

$0 = -1$

The system has no solution.

$$\sim \left(\begin{array}{ccc|c} 2 & -3 & 5 & 1 \\ 0 & 11 & -17 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right) \quad [R_3' = R_3 - R_2]$$

↳ Not a pivot

$$\sim \left(\begin{array}{ccc|c} 22 & 0 & 4 & 14 \\ 0 & 11 & -17 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right) \quad [R_1' = 11R_1 + 3R_2]$$

$$\sim \left(\begin{array}{ccc|c} 1 & 0 & \frac{2}{11} & \frac{7}{11} \\ 0 & 1 & -\frac{17}{11} & \frac{1}{11} \\ 0 & 0 & 0 & 0 \end{array}\right) \quad [\text{Reduced Row Echelon Form}]$$

The Original System: $x + \frac{2}{11}z = \frac{7}{11}$

$$y - \frac{17}{11}z = \frac{1}{11}$$

3rd column has no pivot, so z is a free variable. Let, $z = t$.

$$\therefore y = \frac{1}{11} + \frac{17t}{11} = \frac{1+17t}{11}$$

$$\therefore x = \frac{7-2t}{11}.$$

$$\therefore (x, y, z) = \left(\frac{7-2t}{11}, \frac{1+17t}{11}, t\right).$$

Infinitely Many Solution

No Solution	Infinitely Many	Unique Solution
0 = Non Zero	no. of variable) no. of Pivot	no. of variable = no. of Pivot

Problem-5: Solve the system:

$$2x_1 + 2x_2 + 2x_3 = 0$$
$$-2x_1 + 5x_2 + 2x_3 = 1$$
$$8x_1 + x_2 + 4x_3 = -1$$

$$\therefore \sim \left(\begin{array}{ccc|c} 2 & 2 & 2 & 0 \\ -2 & 5 & 2 & 1 \\ 8 & 1 & 4 & -1 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} 2 & 2 & 2 & 0 \\ 0 & 7 & 4 & 1 \\ 0 & -7 & -4 & -1 \end{array}\right) \quad \left[\begin{array}{l} R_2' = R_2 + R_1 \\ R_3' = R_3 - 4R_1 \end{array}\right]$$

$$\sim \left(\begin{array}{ccc|c} 2 & 0 & 2 & 0 \\ 0 & 7 & 4 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right) \quad \left[R_3' = R_3 + R_2\right]$$

$$\sim \left(\begin{array}{ccc|c} 14 & 0 & 6 & -2 \\ 0 & 7 & 4 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right) \quad \left[R_1' = 7R_1 - 2R_2\right]$$

CamScanner

$$\sim \left(\begin{array}{ccc|c} 14 & 0 & 6 & -2 \\ 0 & 7 & 4 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} 1 & 0 & \frac{3}{7} & -\frac{1}{7} \\ 0 & 1 & \frac{4}{7} & \frac{1}{7} \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$$x + \frac{3}{7}z = -\frac{1}{7}$$

$$y + \frac{4}{7}z = \frac{1}{7}$$

· 3rd Column has no pivot, so z is free variable

Let, $z = t$.

$$\therefore (x, y, z) \equiv \left(\frac{-1-3t}{7}, \frac{1-4t}{7}, t\right) \quad [t \in R]$$

Infinitely many solution.

H.W.: Solve the System:

$$x - y + 2z - w = -1$$
$$2x + y - 2z - 2w = -2$$
$$-x + 2y - 4z + w = 1$$
$$-3w = -3$$
$$3x$$

Q. Find the row picture of the system:

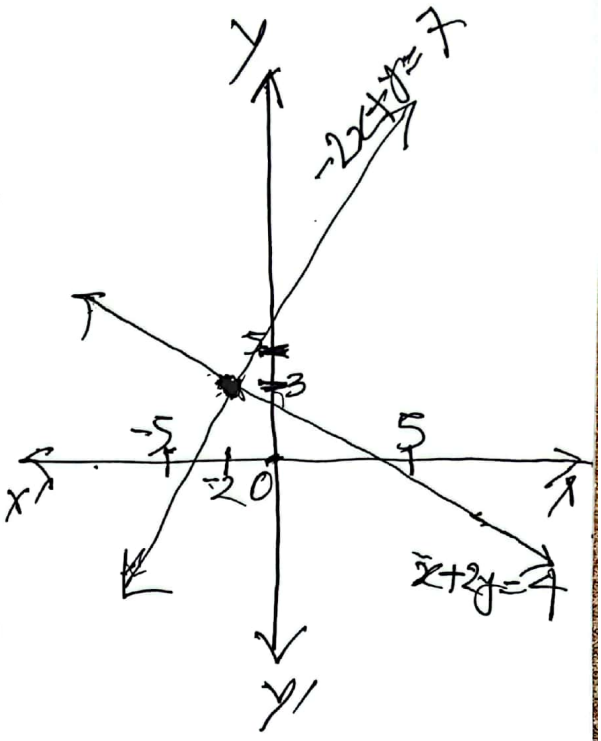
$$\begin{cases} x + 2y = 4 \\ -2x + y = 7 \end{cases}$$

$x + 2y = 4$

x	y
2	1
0	2
−2	3
6	−1

$-2x + y = 7$

x	y
1	9
−1	5
−2	3

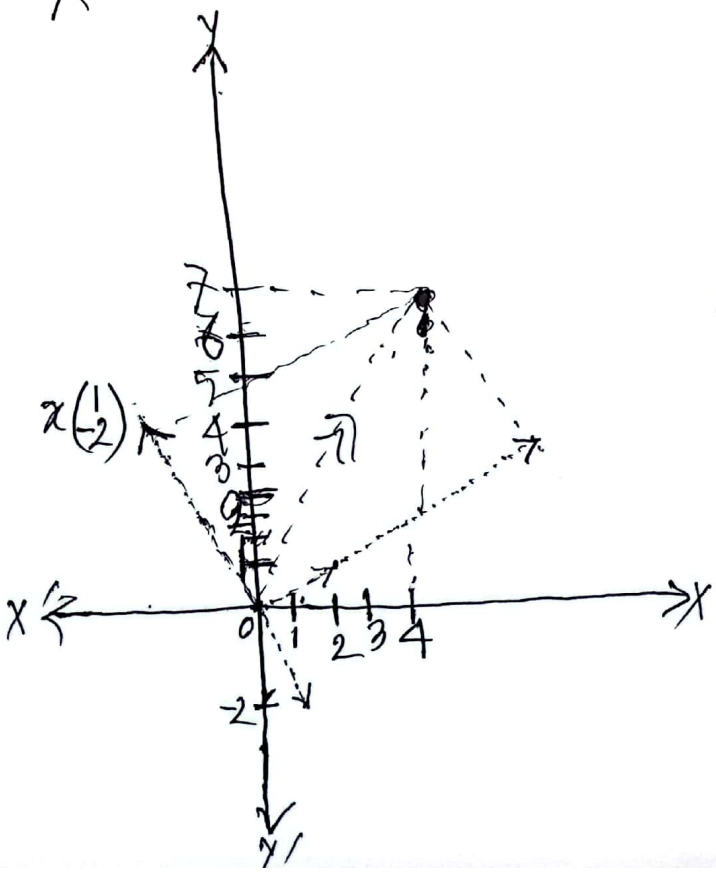


$(x, y) = (-2, 3)$

Q.2 Find the column Picture of the system:

$$\begin{cases} x + 2y = 4 \\ -2x + y = 7 \end{cases}$$

$$\begin{pmatrix} 1 \\ -2 \end{pmatrix} x + \begin{pmatrix} 2 \\ 1 \end{pmatrix} y = \begin{pmatrix} 4 \\ 7 \end{pmatrix}$$



Q. Find the Matrix Picture of the system.

$$\begin{cases} x+2y=4 \\ -2x+y=7 \end{cases}$$

Matrix Picture:

$$\begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}=\begin{pmatrix} 4 \\ 7 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix}=\begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}^{-1}\begin{pmatrix} 4 \\ 7 \end{pmatrix}.$$

Q. Determine the values of λ such that the system of linear equation has:

(i) No solution (ii) More than one solution

(iii) A unique solution.

$$x+y-z=1$$
$$2x+3y+\lambda z=3$$
$$x+\lambda y+3z=2$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 2 & 3 & \lambda & 3 \\ 1 & \lambda & 3 & 2 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & 1 & \lambda+2 & 1 \\ 0 & \lambda-1 & 4 & 1 \end{array}\right) \quad \begin{bmatrix} R_2' = R_2 - 2R_1 \\ R_3' = R_3 - R_1 \end{bmatrix}$$

CamScanner

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & 1 & \lambda+2 & 1 \\ 0 & \lambda-1 & 4 & 1 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & 1 & \lambda+2 & 1 \\ 0 & 0 & 6-\lambda-\lambda^2 & 2-\lambda \end{array}\right) \quad [R_3' = R_3 - (\lambda-1)R_2]$$

$$\left[\begin{array}{l} 4-(\lambda-1)(\lambda+2) \\ = 4-(\lambda^2+\lambda-2) \\ = 6-\lambda-\lambda^2 \end{array}\right]$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & 1 & \lambda+2 & 1 \\ 0 & 0 & +(\lambda+3)\cancel{(\lambda+2)}^{(2-\lambda)} & 2-\lambda \end{array}\right)$$

i) No Solution: $0 = \text{Non zero}$; $\lambda = -3$.

ii) Unique Solution: $\lambda \neq -3$ and $\lambda \neq 2$

iii) Many Solution: $\lambda = 2$.

CamScanner

9. Determine the values of λ and μ such that the system of linear equation has

(i) No solution (ii) More than one solution.

(iii) A unique solution.

$$x+y+z=6$$
$$x+2y+3z=10$$
$$x+2y+\lambda z=\mu$$

$$\sim\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{array}\right)$$

$$\sim\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda-1 & \mu-6 \end{array}\right) \quad \begin{bmatrix} R_2' = R_2 - R_1 \\ R_3' = R_3 - R_1 \end{bmatrix}$$

$$\sim\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda-3 & \mu-10 \end{array}\right) \quad \left[R_3' = R_3 - R_2\right]$$

i) No Solution: $\lambda = 3, \mu \neq 10$

ii) Unique: $\lambda \neq 3$.

iii) Many: $\lambda = 3, \mu = 10$.

CamScanner

Vector Space, Subspace

$\mathbb{R}$ ⟶ Set of all reals.
↳ Field

$\mathbb{R}^2 \rightarrow \{(a,b): a \in \mathbb{R}, b \in \mathbb{R}\} \rightarrow (1,5), (-2,10), (\frac{1}{2}, \frac{3}{2})$

$\mathbb{R}^3 \rightarrow \{(a,b,c): a,b,c \in \mathbb{R}\} \rightarrow (1,0,2), (-1,6,5), (3, \frac{1}{2}, -\frac{2}{3})$.

Formal definition of Vector Space:

Vector Space: A set $\vec{V}$ equipped with two binary ~~oppe~~ operations addition and scalar multiplication is called a vector space over the field $\vec{F}$ if $\vec{V}$ satisfies the following 10 axioms.

1) (A_1) Closed under addition: $u \oplus v \in V$ for all, $u, v \in V$.

2) (A_2) Commutativity: $u \oplus v = v \oplus u$.

3) (A_3) Associativity: $u \oplus (v \oplus w) = (u \oplus v) \oplus w$.

4) (A_4) Additive identity: There exists a $\vec{0} \in V$ such that $u \oplus \vec{0} = u$ for all $u \in V$.

5) (A_5) Additive inverse: There exists $u' \in V$ for all $u \in V$ such that $u \oplus u' = \vec{0}$.

CamScanner

6) (M_1) Closed under Scalar Multiplication (M_1):
$k \cdot u \in V$ for all $u \in V$ and $k \in F$

7) (M_2) Associativity of scalar: $(a \cdot b) \cdot u = a \cdot (b \cdot u)$
for all $a, b \in F$ and $u \in V$.

8) (M_3) Distributive Law: $a \cdot (u \oplus v) = a \cdot u \oplus a \cdot v$
for all $a \in F$ and $u, v \in V$.

9) (M_4) Distributive law: $(a \oplus b) u = a \cdot u \oplus b \cdot u$.
for all $a, b \in F$ and $u \in V$.

10) (M_5) Unit of Scalar Multiplication:
$1 \cdot u = u$. where $1 \in F$ and for all $u \in V$.

~~Problem:~~ $V \rightarrow$ Set of vectors
$F \rightarrow$ Set of Reals.

add : Vectors + Vector = Vector
Multiply: Real . Vector = Vector

Q.1: Consider the set $V = R^2$. A generic element of $\mathbb{R}^2$ is given by the pair of (x, y) where $x, y \in \mathbb{R}$. The operations addition and scalar multiplication is defined on $\mathbb{R}^2$ by $(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$ and $k(x_1, y_1) = (kx_1, 0)$ respectively. Is the set

$\mathbb{R}^2$ with the above defined operations a vector space over the field $F=\mathbb{R}$?

M-5: $1 \cdot u = u$.

Let, $u \equiv (2,5)$

L.S.	R.S.	
$= 1.u$	$= u$	M_5 Fails.
$= 1.(2,5)$	$= (2,5)$	No Vector Space.
$= (2,0)$		

Problems: Determine which sets are vector spaces under the given operations. For those that are not vector spaces, list at least one axiom that fail to hold.

1) The set of all triples of real numbers (x,y,z) with the operations $(x,y,z)+(x',y',z') = (x+x', y+y', z+z')$ and $k(x,y,z) = (0,0,0)$.

2) The set of all pairs of real numbers of the form (x,y), where $x \geq 0$ with the standard operation on $\mathbb{R}^2$.

3) The set of all pairs of real numbers (x,y) with the operations $(x,y)+(x',y') = (x+x'+1, y+y'+1)$ and $k(x,y) = (kx, ky)$.

4) The set of all pairs of real numbers of the form $(1,y)$ with the

$(1,y)+(1,y') = (1, y+y')$ and $k(1,y) = (1,ky)$.

Q.1. $(x,y,z)+(x',y',z') = (x+x', y+y', z+z')$

$k(x,y,z) = (0,0,0)$

Sol^n: Fail at M_5: Let, $u = (2,3,4)$

L.H.S. $= 1 \cdot u$
$= 1 \cdot (2,3,4)$
$= (0,0,0)$

R.S. $= u = (2,3,4)$
L.S. $\neq$ R.S.
No vector space

Q.2. $V = \{(x,y) : x, y \in \mathbb{R}, x \geq 0\}$

$(x,y)+(x',y') = (x+x', y+y')$

$k(x,y) = (kx, ky)$.

Sol^n: $u = (2,5) \in V$, $k = -3 \in \mathbb{R}$ or F

$ku = -3(2,5) = (-6,-15) \notin V$ [M_1 Fails] [No vector space]

Q.3. $(x,y)+(x',y') = (x+x'+1, y+y'+1)$

$k(x,y) = (kx, ky)$

L.S. $= k(u+v)$
$= 4((2,3)+(5,7))$
$= 4(8,11)$
$= (32,44)$

R.S. $= ku+kv$
$= 4(2,3)+4(5,7)$
$= (8,12)+(20,28)$
$= (29,41)$

$u = (2,3)$
$v = (5,7)$
$k = 4$

Fails at
No vector space.

CamScanner

Q.4: $(1,y)+(1,y')=(1,y+y')$ and

$k(1,y)=(1,ky)$

$V=\{(1,y): y\in\mathbb{R}\}$

$(1,y)+(1,y')=(1,y+y')$ ∴

$k(1,y)=(1,ky)$

$(1,y)+(1,0)=(1,y)$

M3 Proof: a $\quad u=(1,x);\ v=(1,x')$

L.H.S. $=a(u+v)$

$=a((1,x)+(1,x'))$

$=a(1,x+x')$

$=(a,ax+ax')$

R.H.S. $=au+av$

$=a(1,x)+a(1,x')$

$=(1,ax)+(1,ax')$

$=(1,ax+ax')$

Q 5: Let $\mathbb{R}^+$ be the set of all positive real numbers. Define vector addition and scalar multiplication by:

$x\oplus y=xy$ for all $x,y\in\mathbb{R}^+$.

$k\cdot x=x^k$ for all $x\in\mathbb{R}^+$ and $k\in\mathbb{R}$.

Show that $\mathbb{R}^+$ is a vector space over the field $\mathbb{R}$.

CamScanner

Q.6. Check whether the set $V=\mathbb{R}$ (set of real numbers) with the usual addition and scalar multiplication is a vector space or not over the field where.

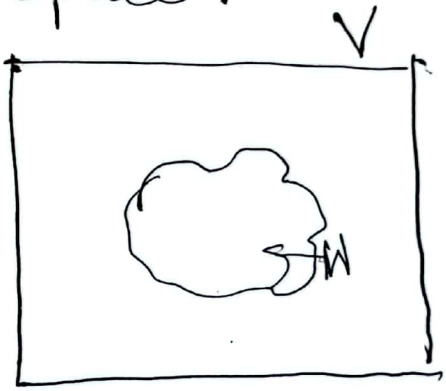
(i) $F=\mathbb{R}$ (set of real numbers)

(ii) $F=\mathbb{Q}$ (set of rational numbers)

(iii) $F=\mathbb{C}$ (set of complex numbers).

Subspaces: Let, V be a Vector Space. A subset W of V is called a subspace of V if W itself a Vector Space.

$W \subseteq V$



Fundamental Theorem of Subspace:

A subset S of a Vector Space V is called a Subspace V if and only if:

(i) A_4 : $\vec{0} \in S$

(ii) A_1 : for any $\vec{u}, \vec{v} \in S \Rightarrow \vec{u}+\vec{v} \in S$.

(iii) M_1 : for any scalar k and any $\vec{u} \in S$
$\Rightarrow k\vec{u} \in S$.

Q: Determine which of the following are subspace of the vector space V.

(a) All vectors of the form $(a,0,0)$, where $V=\mathbb{R}^3$

$S=\{(a,0,0) : a\in\mathbb{R}\}$

A_4: $(x,0,0)+(0,0,0)=(x,0,0)$

A_1: $(x,0,0)+(x',0,0)=(x+x',0,0)\in S$

M_1: $k(x,0,0)=(kx,0,0)\in S$.

Q: Determine which of the following are subspace of the vector space V.

(b) all vectors of the form (a,b,c) with $c=a-b$ where $V=\mathbb{R}^3$

$S=\{(a,b,c) : a,b,c\in\mathbb{R},\ c=a-b\}$

A_4: $(a,b,c)+(0,0,0)=(a,b,c)$

A_1: Let, $(a,b,c)\in S$ when $c=a-b$
$(a',b',c')\in S$ when $c'=a'-b'$

$(a,b,c)+(a',b',c')=(a+a',b+b',c+c')$

$(a+a')-(b+b')=(a-b)+(a'-b')=c-c'$

M_1: Let, $(a,b,c)\in S$ when $c=a-b$.
Let, $k\in F$

$k(a,b,c)=(ka,kb,kc)$

$ka-kb=k(a-b)=kc$

S is a subspace of V.

Q. Determine which of the following are subspace of the vector V.

(c) All vectors of the form (a,b,c) with $c = a+b+3$; where $V = \mathbb{R}^3$.

$S = \{(a,b,c) : c = a+b+3 \; ; \; a,b,c \in \mathbb{R}\}$

Fails at A_4:

$$(a,b,c) + (0,0,0) = (a,b,c)$$

$$0 = 0+0+3 \; ??$$

Q. Determine which of the following are subspace of the vector space V.

(d) All matrices $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ with

$a+b+c+d = 0$, where $V = M_{2\times 2}$.

$S = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a+b+c+d = 0 \right\}$

Proof: A_4: $\begin{pmatrix} a & b \\ c & d \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

A_1: $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ $\quad a+b+c+d = 0$

$\begin{pmatrix} x & y \\ z & w \end{pmatrix}$ $\quad x+y+z+w = 0$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} + \begin{pmatrix} x & y \\ z & w \end{pmatrix} = \begin{pmatrix} a+x & b+y \\ c+z & d+w \end{pmatrix} \in S$$

CamScanner

$$(a+x)+(b+y)+(c+z)+(d+w)$$
$$=(a+b+c+d)+(x+y+z+w)$$
$$=0+0=0$$

H.W.Q. All vectors of the form $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ with $x+2y+3z=5$, where the total vector space $V=R^3$ over the scalar field $F=R$, with usual addition and scalar multiplication.

Q. Consider the vector space $V=M^{2\times 2}$ over the field $\mathbb{R}$ with usual matrix addition and scalar multiplication, where $M^{2\times 2}$ is the set of all 2×2 matrices. Now consider a subset consisting of all 2×2 symmetric matrices of the vector space $M^{2\times 2}$. Is this subset a subspace over the field $\mathbb{R}$? (Hints: for symmetric matrix $A=A^T$.)

$S=\{$all 2×2 symmetric matrix$\}$

$$=\left\{\begin{pmatrix} a & b \\ b & c \end{pmatrix} : a,b,c\in\mathbb{R}\right\}$$

CamScanner

Proof: A_4: $\begin{pmatrix} a & b \\ b & c \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ ✓

A_1: $\begin{pmatrix} a & b \\ b & c \end{pmatrix} + \begin{pmatrix} a' & b' \\ b' & c' \end{pmatrix}$

$= \begin{pmatrix} a+a' & b+b' \\ b+b' & c+d \end{pmatrix}$

$A^T = A$ symmetric: $S = \left\{ \begin{pmatrix} a & b \\ b & c \end{pmatrix} : a, b, c \in \mathbb{R} \right\}$

$A^T = -A$ skew-n: $S = \left\{ \begin{pmatrix} 0 & a \\ -a & 0 \end{pmatrix} : a \in \mathbb{R} \right\}$

CamScanner

Basis & Dimension

Linear Combination & Dependence

Linear Combination: Let V be a vector space over the field F. If $v_1, v_2, v_3, \ldots, v_n \in V$ and $\alpha_1, \alpha_2, \alpha_3, \ldots, \alpha_n \in F$ then $\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 + \ldots + \alpha_n v_n$ is called linear combination.

Q. In the vector space $R_{2\times 2}$, express $A = \begin{bmatrix} 5 & 1 \\ -2 & 3 \end{bmatrix}$ as a linear combination of the matrices $A_1 = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$, $A_2 = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$ and $A_3 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$.

$$A = \alpha_1 A_1 + \alpha_2 A_2 + \alpha_3 A_3$$

$$\begin{pmatrix} 5 & 1 \\ -2 & 3 \end{pmatrix} = \alpha_1 \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} + \alpha_2 \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} \alpha_1 & 0 \\ \alpha_1 & 0 \end{pmatrix} + \begin{pmatrix} \alpha_2 & -\alpha_2 \\ 0 & \alpha_2 \end{pmatrix} + \begin{pmatrix} 0 & \alpha_3 \\ -\alpha_3 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 5 & 1 \\ -2 & 3 \end{pmatrix} = \begin{pmatrix} \alpha_1 + \alpha_2 & -\alpha_2 + \alpha_3 \\ \alpha_1 - \alpha_3 & \alpha_2 \end{pmatrix}$$

$\alpha_1 + \alpha_2 = 5 \Rightarrow \alpha_1 = 2$

$1 = -\alpha_2 + \alpha_3$

$-2 = \alpha_1 - \alpha_3 \Rightarrow \alpha_3 = \alpha_1 + 2 = 4$

$\alpha_2 = 3$

$\therefore A = 2A_1 + 3A_2 + 4A_3$

CamScanner

Q. In the vector space $\mathbb{R}^3$ express the vector $v=(6,-7,3)$ as a linear combination of the vectors $v_1=(1,-2,1)$, $v_2=(2,1,-1)$ and $v_3=(7,-4,1)$.

$$(6,-7,3) = \alpha_1(1,-2,1)+\alpha_2(2,1,-1)+\alpha_3(7,-4,1)$$
$$= (\alpha_1+2\alpha_2+7\alpha_3, -2\alpha_1+\alpha_2-4\alpha_3, \alpha_1-\alpha_2+\alpha_3)$$

$$\left.\begin{aligned} \alpha_1+2\alpha_2+7\alpha_3 &= 6 \\ -2\alpha_1+\alpha_2-4\alpha_3 &= -7 \\ \alpha_1-\alpha_2+\alpha_3 &= 3 \end{aligned}\right]$$

Q. Express $v=(3,5)$ as a linear combination of $(1,2)$ and $(2,4)$.

$$(3,5) = \alpha_1(1,2)+\alpha_2(2,4)$$
$$\Rightarrow (3,5) = (\alpha_1+2\alpha_2, 2\alpha_1+4\alpha_2)$$

$$\left.\begin{aligned} \alpha_1+2\alpha_2 &= 3 \\ 2\alpha_1+4\alpha_2 &= 5 \end{aligned}\right|$$

$$\left(\begin{array}{cc|c} 1 & 2 & 3 \\ 2 & 4 & 5 \end{array}\right)$$

$$\sim \left(\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 0 & -1 \end{array}\right) \quad [R_2' = R_2 - 2R_1]$$

↳ No solution

So, it's not possible to express v as a linear combination of other vectors.

Linear Dependence:

$S=\{v_1, v_2, v_3, \ldots, v_n\}$ the set of vectors is called dependent if the system

$$\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 + \cdots + \alpha_n v_n = 0$$

has many solution.

Q. Show that the set of vectors $\{(2,1,2),(0,1,-1),(4,3,3)\}$ in $\mathbb{R}^3$ is linearly dependent.

$$\alpha_1(2,1,2)+\alpha_2(0,1,-1)+\alpha_3(4,3,3)=(0,0,0)$$

$$\Rightarrow (2\alpha_1+4\alpha_3, \alpha_1+\alpha_2+3\alpha_3, 2\alpha_1-\alpha_2+3\alpha_3)=(0,0,0)$$

$$2\alpha_1+4\alpha_3=0$$
$$\alpha_1+\alpha_2+3\alpha_3=0$$
$$2\alpha_1-\alpha_2+3\alpha_3=0$$

$$\left(\begin{array}{ccc|c} 2 & 0 & 4 & 0 \\ 1 & 1 & 3 & 0 \\ 2 & -1 & 3 & 0 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} 2 & 0 & 4 & 0 \\ 0 & 2 & 2 & 0 \\ 0 & -1 & -1 & 0 \end{array}\right) \begin{array}{l} R_2'=2R_2-R_1 \\ R_3'=R_3-R_1 \end{array}$$

$$\sim \left(\begin{array}{ccc|c} 2 & 0 & 4 & 0 \\ 0 & 2 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \begin{array}{l} \\ \\ R_3'=2R_3+R_2 \end{array}$$

The system has many solution.

9. Show that the set of vectors $\{(3,0,1,-1),(2,-1,0,1),(1,1,1,-2)\}$ in $\mathbb{R}^4$ is linearly dependent.

$$\alpha_1(3,0,1,-1)+\alpha_2(2,-1,0,1)+\alpha_3(1,1,1,-2)=(0,0,0,0)$$

$$(3\alpha_1+2\alpha_2+\alpha_3,\ -\alpha_2+\alpha_3,\ \alpha_1+\alpha_3,\ -\alpha_1+\alpha_2-2\alpha_3)=(0,0,0,0)$$

$$\therefore \begin{aligned} 3\alpha_1+2\alpha_2+\alpha_3&=0\\ -\alpha_2+\alpha_3&=0\\ \alpha_1+\alpha_3&=0\\ -\alpha_1+\alpha_2-2\alpha_3&=0 \end{aligned}$$

$$\left(\begin{array}{ccc|c} 3 & 2 & 1 & 0\\ 0 & -1 & 1 & 0\\ 1 & 0 & 1 & 0\\ -1 & 1 & -2 & 0 \end{array}\right)$$

CamScanner

Q. Show that the set of vectors $\{(2,1,2), (0,1,-1), (4,3,4)\}$ in $\mathbb{R}^3$ is linearly independent.

$\alpha_1(2,1,2) + \alpha_2(0,1,-1) + \alpha_3(4,3,4) = (0,0,0)$

$$\left.\begin{aligned} 2\alpha_1 + 4\alpha_3 &= 0 \\ \alpha_1 + \alpha_2 + 3\alpha_3 &= 0 \\ 2\alpha_1 - \alpha_2 + 4\alpha_3 &= 0 \end{aligned}\right\} \qquad \left(\begin{array}{ccc|c} 2 & 0 & 4 & 0 \\ 1 & 1 & 3 & 0 \\ 2 & -1 & 4 & 0 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} 2 & 0 & 4 & 0 \\ 0 & 2 & 2 & 0 \\ 0 & -1 & 0 & 0 \end{array}\right) \begin{bmatrix} R_2' = 2R_2 - R_1 \\ R_3' = R_3 - R_1 \end{bmatrix}$$

$$\sim \left(\begin{array}{ccc|c} 2 & 0 & 4 & 0 \\ 0 & 2 & 2 & 0 \\ 0 & 0 & 2 & 0 \end{array}\right) \begin{bmatrix} R_3' = 2R_3 + R_2 \end{bmatrix}$$

$\Rightarrow$ Unique solution $\rightarrow$ Independent.

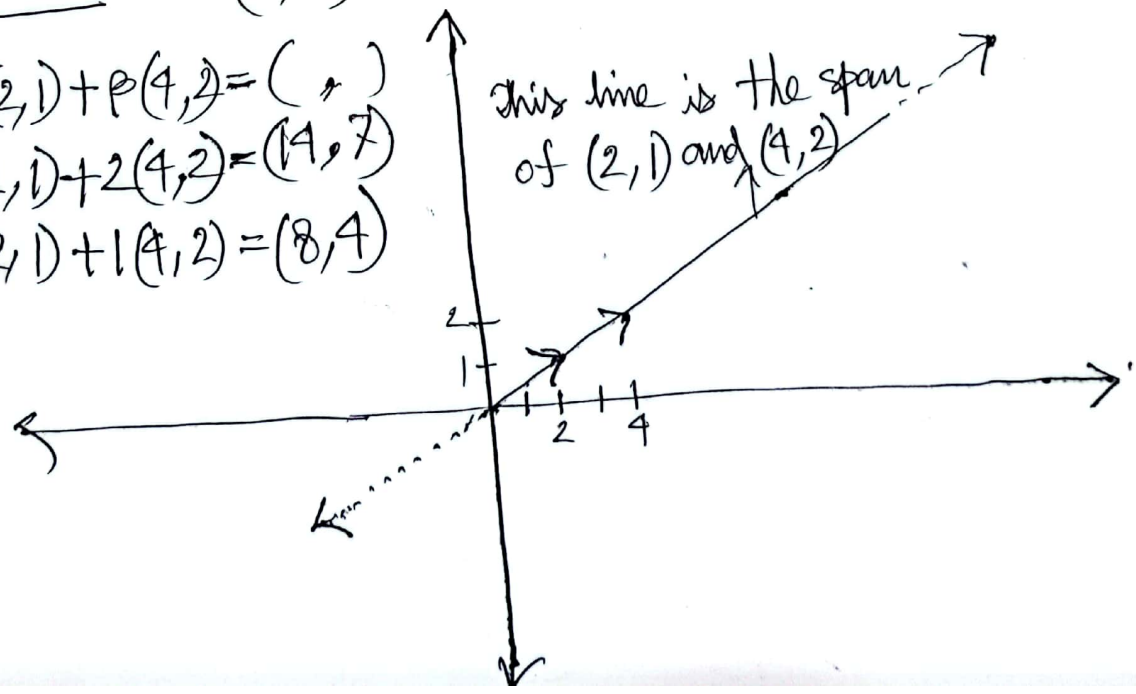
Basis and Dimension

Span: ~~$(1,2)$ and $(2,4)$~~ $(2,1)$ and $(4,2)$

$\alpha(2,1) + \beta(4,2) = (x, y)$

$3(2,1) + 2(4,2) = (14, 7)$

$2(2,1) + 1(4,2) = (8, 4)$



Q. What is the span of (2,1) and (4,2)?

span $\{(2,1),(4,2)\}$ = line $x=2y$

$=\{(x,y): x,y\in\mathbb{R}, x=2y\}$

Q. Show that the vectors (1,2) and (2,1) span the whole $\mathbb{R}^2$.

Consider a vector (a,b) in $\mathbb{R}^2$.

$\alpha_1(1,2)+\alpha_2(2,1)=(a,b)$

$\alpha_1+2\alpha_2=a$
$2\alpha_1+\alpha_2=b$

$$\left(\begin{array}{cc|c}1 & 2 & a\\ 2 & 1 & b\end{array}\right) \sim \left(\begin{array}{cc|c}1 & 2 & a\\ 0 & -3 & b-2a\end{array}\right) \rightarrow \text{unique.}$$

$\alpha_1(2,1)+\alpha_2(4,2)=(a,b)$

$2\alpha_1+4\alpha_2=a$
$\alpha_1+2\alpha_2=b$

$$\left(\begin{array}{cc|c}2 & 4 & a\\ 1 & 2 & b\end{array}\right) \sim \left(\begin{array}{cc|c}2 & 4 & a\\ 0 & 0 & 2b-a\end{array}\right)$$

→ Soln if $a=2b$

→ No solution. if $2b\neq a$.

Q. Show that the vectors (1,2), (2,4) and (3,2) span the whole $\mathbb{R}^2$.

$\alpha_1(1,2)+\alpha_2(2,4)+\alpha_3(3,2)=(a,b)$

$\alpha_1+2\alpha_2+3\alpha_3=a$
$2\alpha_1+4\alpha_2+2\alpha_3=b$

$$\left(\begin{array}{ccc|c}1 & 2 & 3 & a\\ 2 & 4 & 2 & b\end{array}\right) \sim \left(\begin{array}{ccc|c}1 & 2 & 3 & a\\ 0 & 0 & -4 & b-2a\end{array}\right)$$

Basis: A set of independent vectors that spans all the vectors of a vector space.

Dimension: Number of vectors in Basis

Four Fundamental Spaces:

Row space, Column space, Null space, Left Null

Row Space and Row Rank:

- $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 2 & 8 \\ 6 & 2 & 9 \end{pmatrix}$

span{(1,2,3), (4,2,8), (6,2,9)}

$\sim \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{pmatrix}$ $\sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

→ span{(1,2,3), (2,4,6), (3,6,9)}

→ span{(1,2,3), (2,4,6)}

→ span{(1,2,3)}

9. Consider the matrix of a transformation

$$\begin{pmatrix} 2 & 1 & 1 \\ -1 & 2 & 7 \\ 1 & -1 & -4 \end{pmatrix}$$

Find the Row Space of T.

CamScanner

Soln.
$$\begin{pmatrix} 2 & 1 & 1 \\ -1 & 2 & 7 \\ 1 & -1 & -4 \end{pmatrix}$$
$$\sim \begin{pmatrix} 2 & 1 & 1 \\ 0 & 5 & 15 \\ 0 & -3 & -9 \end{pmatrix}$$
$$\sim \begin{pmatrix} 2 & 1 & 1 \\ 0 & 5 & 15 \\ 0 & 0 & 0 \end{pmatrix}$$

Row space = Span$\{(2,1,1), (-1,2,7), (1,-1,4)\}$
= span$\{(2,1,1), (-1,2,7)\}$

Basis of Row Space = $\{(2,1,1), (-1,2,7)\}$

Dimension = 2.

Q. Consider the matrix of transformation.
$$\begin{pmatrix} 1 & 1 & 3 \\ -2 & 5 & 1 \\ 1 & 3 & 5 \\ 3 & -2 & 4 \end{pmatrix}$$

Find the row space of T.

$$\begin{pmatrix} 1 & 1 & 3 \\ -2 & 5 & 1 \\ 1 & 3 & 5 \\ 3 & -2 & 4 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 3 \\ 0 & 7 & 7 \\ 0 & 2 & 2 \\ 0 & -5 & -5 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 3 \\ 0 & 7 & 7 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Row space = Span$\{(1,1,3), (-2,5,1), (1,3,5), (3,-2,4)\}$
= span$\{(1,1,3), (-2,5,1)\}$

Basis of Row space = $\{(1,1,3), (-2,5,1)\}$ rank = 2.

9. Consider the matrix of transformation:

$$\begin{pmatrix} 0 & -2 & 1 & 3 \\ 1 & 5 & 3 & -2 \\ 1 & 1 & 5 & 4 \end{pmatrix}.$$

Find the row space of T.

Soln:

$$\begin{pmatrix} 0 & -2 & 1 & 3 \\ 1 & 5 & 3 & -2 \\ 1 & 1 & 5 & 4 \end{pmatrix}$$

$$\sim \begin{pmatrix} 1 & 5 & 3 & -2 \\ 0 & -2 & 1 & 3 \\ 1 & 1 & 5 & 4 \end{pmatrix} [R_1 \leftrightarrow R_2]$$

$$\sim \begin{pmatrix} 1 & 5 & 3 & -2 \\ 0 & -2 & 1 & 3 \\ 0 & -4 & 2 & 6 \end{pmatrix} [R_3' = R_3 - R_1]$$

$$\sim \begin{pmatrix} 1 & 5 & 3 & -2 \\ 0 & -2 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} [R_3' = R_3 - 2R_2]$$

Row Space = span $\{(0,-2,1,3), (1,5,3,-2), (1,1,5,4)\}$

= span $\{(1,5,3,-2), (1,1,5,4)\}$

Basis = $\{(1,5,3,-2), (1,1,5,4)\}$

CamScanner

Q. Consider the matrix of a transformation:

$$A=\begin{pmatrix}2 & 1 & 1\\ -1 & 2 & 7\\ 1 & -1 & -4\end{pmatrix}$$

Find the column space of T.

Sol^n: Column space of A = Row space of A^T.

$$A^T=\begin{pmatrix}2 & -1 & 1\\ 1 & 2 & -1\\ 1 & 7 & -4\end{pmatrix}$$

Row space of A^T = span$\{(2,-1,1),(1,2,-1)\}$

column space of A = span$\left\{\begin{pmatrix}2\\-1\\1\end{pmatrix},\begin{pmatrix}1\\2\\-1\end{pmatrix}\right\}$

$$\sim\begin{pmatrix}2 & -1 & +1\\ 0 & 5 & -3\\ 0 & 15 & -9\end{pmatrix}$$

Dimension = 2

Column Rank = 2.

$$\sim\begin{pmatrix}2 & -1 & 1\\ 0 & 5 & -3\\ 0 & 0 & 0\end{pmatrix}$$

Q: Consider the matrix of transformation:

$$A=\begin{pmatrix}0 & -2 & 1 & 3\\ 1 & 5 & 3 & -2\\ 1 & 1 & 5 & 4\end{pmatrix}$$

Find the column space of T.

$$A^T=\begin{pmatrix}0 & 1 & 1\\ -2 & 5 & 1\\ 1 & 3 & 5\\ 3 & -2 & 4\end{pmatrix}$$

CamScanner

$$A^T = \begin{pmatrix} 0 & 1 & 1 \\ -2 & 5 & 1 \\ 1 & 3 & 5 \\ 3 & -2 & 4 \end{pmatrix}$$

$$\sim \begin{pmatrix} 1 & 3 & 5 \\ -2 & 5 & 1 \\ 0 & 1 & 1 \\ 3 & -2 & 4 \end{pmatrix} \quad [R_1 \leftrightarrow R_3]$$

$$\sim \begin{pmatrix} 1 & 3 & 5 \\ 0 & 11 & 11 \\ 0 & 1 & 1 \\ 0 & -11 & -11 \end{pmatrix} \quad \begin{matrix} R_2' = R_2 + 2R_1 \\ R_4' = R_4 - 3R_1 \end{matrix}$$

$$\therefore \sim \begin{pmatrix} 1 & 3 & 5 \\ 0 & 11 & 11 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \begin{matrix} R_3' = 11R_3 - R_2 \\ R_4' = R_4 + R_2 \end{matrix}$$

Row space of $A^T = \text{span}\{(-2,5,1), (1,3,5)\}$

Column space of $A = \text{span}\left\{\begin{pmatrix} -2 \\ 5 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix}\right\}$

$$\text{Basis} = \left\{\begin{pmatrix} -2 \\ 5 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix}\right\}$$

Dimension = 2.

Column rank = 2.

CamScanner

Null Space and Nullity:

$$\begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x+4y \\ x+2y \end{pmatrix}$$

$$\begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Null space of $A = \{x : Ax = 0\}$

Q. Consider the matrix of transformation

$$\begin{pmatrix} 2 & -1 & 1 \\ 1 & 2 & -1 \\ 1 & 7 & -4 \end{pmatrix}.$$

Find the Null Space (T).

Consider, $\begin{pmatrix} x \\ y \\ z \end{pmatrix} \in$ Null space

$$\begin{pmatrix} 2 & -1 & 1 \\ 1 & 2 & -1 \\ 1 & 7 & -4 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \left(\begin{array}{ccc|c} 2 & -1 & 1 & 0 \\ 1 & 2 & -1 & 0 \\ 1 & 7 & -4 & 0 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} 2 & -1 & 1 & 0 \\ 0 & 5 & -3 & 0 \\ 0 & 15 & -9 & 0 \end{array}\right) \quad \begin{bmatrix} R_2' = 2R_2 - R_1 \\ R_3' = 2R_3 - R_1 \end{bmatrix}$$

$$\sim \left(\begin{array}{ccc|c} 2 & -1 & 1 & 0 \\ 0 & 5 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \quad \begin{bmatrix} R_3' = R_3 - 3R_2 \end{bmatrix}$$

$$\sim \left(\begin{array}{ccc|c} 2 & -1 & 1 & 0 \\ 0 & 5 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} 10 & 0 & 2 & 0 \\ 0 & 5 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \quad [R_1 = 5R_1 + R_2]$$

$$\sim \left(\begin{array}{ccc|c} 1 & 0 & 1/5 & 0 \\ 0 & 1 & -3/5 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$$x + \frac{1}{5}z = 0$$

$$y - \frac{3}{5}z = 0$$

$$\begin{cases} x = -\frac{t}{5} \\ y = \frac{3t}{5} \\ z = t \end{cases}$$

z is a free variable, so, $z = t$.

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -\frac{t}{5} \\ \frac{3t}{5} \\ t \end{pmatrix} = t \begin{pmatrix} -1/5 \\ 3/5 \\ 1 \end{pmatrix}$$

Null space $= \text{span}\left\{\begin{pmatrix} -1/5 \\ 3/5 \\ 1 \end{pmatrix}\right\}$

Basis $= \left\{\begin{pmatrix} -1/5 \\ 3/5 \\ 1 \end{pmatrix}\right\}$

Nullity/Dimension $= 1$.

CamScanner